11 Chemistry Unit 2 Topic Test 1: Concentration, Standard Solutions and Dilution — Worked Solutions and Marking Guide

TEACHER MARKING GUIDE — not for distribution to students

Total 40 marks: Section A 10 × 1 mark (separate multiple-choice answer sheet), Section B 30 marks. Writing time 46 minutes (40 marks × 1.15 min per mark). Students use the VCE Chemistry Data Book (SLC 25 °C and 100 kPa; V_m = 24.8 L mol^{-1} at SLC; R = 8.31 J mol^{-1} K^{-1}; 1 atm = 101.3 kPa). **One mark per idea or step; no half marks.** Units and significant figures are marked once, at the final answer of each part. **Rounding tolerance:** accept a final answer that differs only in the last significant figure because intermediate values were rounded; wider bands are stated where needed. **Consequential marking:** where a part uses an earlier answer, award full marks for correct working from the student's own earlier value (shown as *Consequential*). Each item is labelled with its difficulty (Easy / Medium / Hard, by marks: 10 / 20 / 10) and demand level (L1 recall, L2 routine procedure, L3 interpret or explain, L4 evaluate or justify).

Section A — Multiple-choice answer key

Key balance: A × 3, B × 2, C × 2, D × 3 (key string B D A C B A D C A D). Every distractor is the answer produced by one named slip, so tallies of wrong letters show which misconceptions to reteach.

Q	Key	Working for the key	Distractor diagnosis
1	**B**	**Easy (L1).** % (m/v) = grams of solute per 100 mL of solution.	A: used the volume of solvent (water) instead of the volume of solution; C: confused % (m/v) with % (m/m); D: confused % (m/v) with g L^{-1} (out by a factor of 10).
2	**D**	**Easy (L2).** n = 3.40 ÷ 85.0 = 0.0400 mol; c = 0.0400 ÷ 0.2000 = 0.200 M	A: volume left in mL (0.0400 ÷ 200.0); B: gave the g L^{-1} value (3.40 ÷ 0.2000) without converting mass to mol; C: multiplied the amount by the volume in L instead of dividing.
3	**A**	**Easy (L1).** A standard solution has an accurately known concentration (made from a primary standard, or standardised by titration).	B: thinks 'standard' means one fixed concentration; C: confuses a standard solution with a diluted (working) solution; D: confuses 'standard solution' with SLC.
4	**C**	**Medium (L2).** 2.0% (m/v) = 2.0 g per 100 mL = 20 g L^{-1}; M(H_3BO_3) = 3(1.0) + 10.8 + 3(16.0) = 61.8 g mol^{-1}; c = 20 ÷ 61.8 = 0.32 M	A: treated 2.0% (m/v) as 2.0 g L^{-1}; B: multiplied by 100 instead of 10 (200 g L^{-1}); D: multiplied by the molar mass instead of dividing.
5	**B**	**Medium (L3).** Dilution does not change the amount of solute. Final volume = 200 mL; c_2 = 2.0 × 50 ÷ 200 = 0.50% (m/v)	A: used the 150 mL of water added as the final volume (2.0 × 50 ÷ 150); C: thinks adding water removes solute; D: inverted the volume ratio (2.0 × 200 ÷ 50).
6	**A**	**Medium (L2).** $c_1 = c_2V_2 \div V_1$ = 0.0170 × 200.0 ÷ 4.00 = 0.850 M (dilution factor 50)	B: applied the dilution factor the wrong way (0.0170 × 4.00 ÷ 200.0); C: used 196.0 mL, the volume of water added, as the final volume; D: took the final volume as 204.0 mL (200.0 mL of water plus the aliquot).
7	**D**	**Easy (L2).** $Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-}$; [$SO_4^{2-}$] = 3 × 0.0200 = 0.0600 M	A: ignored the subscript 3 (took [SO_4^{2-}] = [$Al_2(SO_4)_3$]); B: used the subscript of Al (2) instead of SO_4 (3); C: counted all the ions (2 + 3 = 5).
8	**C**	**Hard (L2).** M($C_8H_{10}N_4O_2$) = 194.0 g mol^{-1}; m = 0.320 g L^{-1} × 0.500 L = 0.160 g; n = 0.160 ÷ 194.0 = 8.25 × 10^{-4} mol	A: gave the amount in 1 L (did not use the 500 mL volume); B: did not convert mg to g; D: multiplied by 2 instead of halving for 500 mL.
9	**A**	**Medium (L1).** I and II are correct practice. III is wrong: volumetric glassware is calibrated at a set temperature and must never be heated.	B: does not know the solution must be mixed after making up to the mark; C: does not recognise that heating changes the

			calibrated volume of the flask, and rejects dissolving first in a beaker; D: does not recognise that heating changes the calibrated volume of the flask.
10	**D**	**Hard (L3).** $n(NH_4^+) = 0.180 \times 0.2000 = 0.0360$ mol; $n((NH_4)_2SO_4) = 0.0180$ mol; M = 132.1 g mol^{-1}; m = 0.0180 × 132.1 = 2.38 g	A: used the NH_4^+ concentration as the salt concentration (ignored the 2 : 1 ratio); B: used a volume of 2 L instead of 0.2000 L; C: wrote the formula as NH_4SO_4 (M = 114.1 g mol^{-1}).

Section B — Short-answer worked solutions

Each part shows the worked solution, then the mark allocation (one mark per step or idea), consequential notes and answers to reject.

Question 1 — Descaling a coffee machine (7 marks)

a) (1 mark)

c = m ÷ V = 25.0 g ÷ 0.750 L = 33.3 g L^{-1}

1 mark 33.3 g L^{-1} (unit required). *[Easy; L2]*

Reject: 0.0333 g L^{-1} (volume left in mL); 18.8 g L^{-1} (multiplied by 0.750).

b) (1 mark)

33.3 g in 1000 mL = 3.33 g in 100 mL, so 3.33% (m/v)

1 mark 3.33% (m/v) (accept 3.3%). *[Easy; L2]*

Consequential: On part a): answer = (part a) ÷ 10.

Reject: 333% or 0.333% (wrong power of ten).

c) (3 marks)

$M(C_6H_8O_7) = 6(12.0) + 8(1.0) + 7(16.0) = 192.0$ g mol^{-1}

n = 25.0 ÷ 192.0 = 0.130 mol

c = 0.130 ÷ 0.750 = 0.174 M

1 mark $M(C_6H_8O_7) = 192.0$ g mol^{-1}. **1 mark** n = 0.130 mol (or 33.3 g L^{-1} ÷ 192.0 set up). **1 mark** 0.174 M (accept 0.173–0.174). *[Medium; L2]*

Reject: 0.0977 M (multiplied by 0.750); molar-mass errors (M must be 192.0 g mol^{-1}).

d) (2 marks)

$c_1V_1 = c_2V_2$: 0.174 × V_1 = 0.0800 × 500

V_1 = 40.0 ÷ 0.174 = 230 mL

1 mark correct $c_1V_1 = c_2V_2$ set-up with the concentrated solution as c_1. **1 mark** 230 mL (accept 229–231 mL). *[Medium; L2]*

Consequential: On part c): V_1 = 40.0 ÷ (part c).

Reject: 1090 mL (ratio inverted, 0.174 × 500 ÷ 0.0800); 270 mL (500 − 230, the water added).

Parts a) to c) practise the three concentration units on one solution; part d) is a dilution.

Question 2 — Weighing a primary standard (7 marks)

a) (1 mark)

m = 18.734 − 13.822 = 4.912 g

1 mark 4.912 g. *[Easy; L2]*

Reject: 18.734 g or 13.822 g (a single reading); 32.556 g (readings added).

b) (2 marks)

n = 4.912 ÷ 97.1 = 0.05059 mol

c = 0.05059 ÷ 0.5000 = 0.101 M (0.1012 M)

1 mark n = 0.0506 mol. **1 mark** c = 0.101 M (accept 0.1012 M). *[Medium; L2]*

Consequential: On part a).

Reject: 1.01 × 10^{-4} M (volume in mL); 0.0253 M (multiplied by 0.5000).

c) (2 marks)

Solid NaOH absorbs water vapour from the air and reacts with carbon dioxide, CO_2, in the air.

So the mass weighed is not all NaOH: the amount (mol) of NaOH is not accurately known, and the concentration cannot be calculated accurately from the mass.

1 mark NaOH absorbs water from the air (hygroscopic) **or** reacts with CO_2 in the air (not of known, constant composition/purity). **1 mark** so the mass weighed does not give an accurately known amount of NaOH, and hence the concentration is not accurately known. *[Medium; L3]*

Reject: 'NaOH is a strong base'; 'NaOH is dangerous/corrosive' (true but not the reason); 'NaOH does not dissolve' (false).

d) (2 marks)

It does not affect the result.

Both readings are 0.020 g too high, and the mass transferred is found by subtraction: (18.734 − 0.020) − (13.822 − 0.020) = 4.912 g. The offset cancels, so the mass, amount and concentration are unchanged.

1 mark no effect on the calculated concentration (committed judgement). **1 mark** reason: weighing by difference: both readings carry the same offset, which cancels when one is subtracted from the other. *[Hard; L4]*

Reject: 'Concentration too high because the mass is 0.020 g too high' (ignores weighing by difference); 'lower'; a judgement with no reason scores 0.

Part d) rewards understanding of why technicians weigh by difference.

Question 3 — Cleaning brewery tanks (7 marks)

a) (1 mark)

2.00 g per 100 mL = 20.0 g per 1000 mL = 20.0 g L^{-1}

1 mark 20.0 g L^{-1}. *[Easy; L2]*

Reject: 2.00 g L^{-1}; 200 g L^{-1}.

b) (1 mark)

c = 20.0 g L^{-1} ÷ 40.0 g mol^{-1} = 0.500 M

1 mark 0.500 M. *[Easy; L2]*

Consequential: On part a): (part a) ÷ 40.0.

Reject: 800 M (multiplied by 40.0).

c) (2 marks)

$c_1V_1 = c_2V_2$: 30.0 × V_1 = 2.00 × 600

V_1 = 1200 ÷ 30.0 = 40.0 L

1 mark $c_1V_1 = c_2V_2$ set up correctly (any consistent units: % (m/v), g L^{-1} or mol L^{-1}). **1 mark** 40.0 L. *[Medium; L2]*

Reject: 9000 L (ratio inverted); 560 L (water added, not stock).

d) (3 marks)

Mass of NaOH already in the tank = 14.0 g L^{-1} × 450 L = 6300 g (6.30 kg)

Stock = 300 g L^{-1}; wash needed = 20.0 g L^{-1}. Let V = volume of stock added (L):

6300 + 300V = 20.0 × (450 + V)

280V = 9000 − 6300 = 2700, so V = 9.64 L

1 mark mass (or amount) of NaOH already present, 6300 g. **1 mark** equation that equates total NaOH to 2.00% of the new total volume (450 + V). **1 mark** V = 9.64 L (accept 9.6 L). *[Hard; L3]*

Consequential: Allow consequential use of the student's g L^{-1} conversion from part a).

Reject: 9.00 L (ignores the added volume: 2700 ÷ 300); 21.0 L (1.40% × 450 L ÷ 30.0%: $c_1V_1 = c_2V_2$ applied to the 1.40% wash).

Part d) cannot be done with $c_1V_1 = c_2V_2$ alone: students must track the mass of solute already present and the change in total volume.

Question 4 — Zinc sprays for fruit trees (9 marks)

a) (1 mark)

0.600 g per 100 mL = 6.00 g L^{-1}

1 mark 6.00 g L^{-1}. *[Easy; L2]*

Reject: 0.600 g L^{-1}; 60.0 g L^{-1}.

b) (3 marks)

$M(ZnSO_4) = 65.4 + 32.1 + 4(16.0) = 161.5$ g mol^{-1}

B: 7.50 ÷ 161.5 = 0.0464 M

C: 6.00 ÷ 161.5 = 0.0372 M

A (0.0480 M) > B (0.0464 M) > C (0.0372 M), so concentrate A is the most concentrated.

1 mark B = 0.0464 M. **1 mark** C = 0.0372 M. **1 mark** conclusion: A is highest, consistent with the student's values. *[Medium; L3]*

Consequential: Conversion of C consequential on part a). The conclusion mark is consequential on the student's calculated values.

Reject: Choosing B because 7.50 is the largest number on the labels (no conversion); a conclusion with no working scores 0.

c) (2 marks)

$[SO_4^{2-}] = [ZnSO_4] = 0.0464$ M (1 : 1)

$0.0464 \times 96.1 = 4.46$ g L^{-1} (or $7.50 \times 96.1 \div 161.5 = 4.46$ g L^{-1})

1 mark 1 : 1 ratio, 0.0464 mol L^{-1} of SO_4^{2-} (or the mass-fraction method set up). **1 mark** 4.46 g L^{-1} (accept 4.45–4.47). *[Medium; L2]*

Consequential: On the value for B in part b).

Reject: 7.50 g L^{-1} (the salt, not the ion); 3.04 g L^{-1} (zinc ion mass instead of sulfate).

d) (3 marks)

m(Zn^{2+}) needed = 0.500 g L^{-1} × 100 L = 50.0 g

n(Zn^{2+}) = 50.0 ÷ 65.4 = 0.765 mol = $n(ZnSO_4)$

V(A) = n ÷ c = 0.765 ÷ 0.0480 = 15.9 L

1 mark 50.0 g of Zn^{2+}. **1 mark** 0.765 mol (using 65.4 g mol^{-1} for Zn). **1 mark** 15.9 L. *[Hard; L3]*

Reject: 6.45 L (used $M(ZnSO_4) = 161.5$ g mol^{-1} instead of 65.4 for the zinc ions); 0.159 L (used 0.500 g as the mass of Zn^{2+}, forgetting the 100 L).

Part d) needs the ion/compound distinction: the target is set for Zn^{2+}, but the concentrate is labelled as $ZnSO_4$.

Mark summary

Section	Marks	Notes
A — multiple choice	10	10 × 1 mark; key A3 / B2 / C2 / D3
B — short answer	30	Q1 7, Q2 7, Q3 7, Q4 9
Total	**40**	Writing time 46 minutes

Difficulty tally (by marks)

Section	Easy	Medium	Hard	Total
A — multiple choice	4	4	2	10
B — short answer	6	16	8	30
Total	**10**	**20**	**10**	**40**
Target (25 / 50 / 25%)	10	20	10	40

Difficulty and demand by item

Item	Marks	Difficulty	Demand
A Q1	1	Easy	L1
A Q2	1	Easy	L2
A Q3	1	Easy	L1
A Q4	1	Medium	L2
A Q5	1	Medium	L3
A Q6	1	Medium	L2
A Q7	1	Easy	L2
A Q8	1	Hard	L2
A Q9	1	Medium	L1
A Q10	1	Hard	L3
B Q1a)	1	Easy	L2
B Q1b)	1	Easy	L2
B Q1c)	3	Medium	L2
B Q1d)	2	Medium	L2
B Q2a)	1	Easy	L2
B Q2b)	2	Medium	L2
B Q2c)	2	Medium	L3
B Q2d)	2	Hard	L4
B Q3a)	1	Easy	L2
B Q3b)	1	Easy	L2
B Q3c)	2	Medium	L2
B Q3d)	3	Hard	L3
B Q4a)	1	Easy	L2
B Q4b)	3	Medium	L3
B Q4c)	2	Medium	L2
B Q4d)	3	Hard	L3

Cognitive demand

Demand level	Marks	Share
L1 recall	3	8%
L2 routine procedure	22	55%
L3 interpret / explain	13	32%
L4 evaluate / justify	2	5%

Routine marks (L1 + L2): 25 of 40 (62%).

Key knowledge assessed

• concentration in mol L^{-1}, g L^{-1} and % (m/v), and conversions between them

• standard solutions: primary standards and the volumetric flask procedure

• dilution, $c_1V_1 = c_2V_2$

Not assessed (not taught this year): indicators and indicator choice, pH calculations, back titrations, calibration curves, solubility curves, colorimetry, gravimetric analysis.